

Muhammad Zamin

647-764-5530 | zaminjamal.zj@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

York University

Toronto, ON

Specialized Honours B.Sc. in Computer Science

Relevant Coursework: Data Structures & Algorithms, Software Design, Machine Learning, Computer Security, Big Data, Data Mining, User Interface Design, Mobile Development

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Java, Python, SQL, C, Bash

Frontend: React.js, Next.js, Vite, Tailwind CSS, Redux, HTML, shadcn/ui

Backend: Node.js, Express.js, Spring Boot, REST APIs, WebSockets (Socket.IO), Flask

Databases & Auth: PostgreSQL, Supabase, SQLite, JWT, OAuth, MongoDB

Tools & DevOps: Docker, Git, Postman, Unix/Linux, CI/CD, NPM, Spark, Hadoop

EXPERIENCE

Front-End Developer

Sep 2025 – Present

Department of Computer Science, Lassonde School of Engineering, York University

Toronto, ON

- Developed a real-time virtual queue system for efficient handling of student service requests.
- Built a responsive frontend using React.js and Vite with modular components.
- Applied SDLC principles, translating wireframes into structured implementations.
- Implemented authentication, session management, and WebSockets for real-time updates.

Full-Stack Developer

Jan 2026 – Mar 2026

TMA Student Club, York University

Toronto, ON

- Built a full-stack banquet RSVP platform with user registration and admin check-in mechanisms.
- Developed frontend with Next.js, React, TypeScript, and Framer Motion for smooth UX.
- Engineered scalable database architectures with Supabase and secure authentication using Clerk.
- Implemented secure ticket purchasing and automatic email confirmations using Stripe and Resend.
- Built an admin dashboard with integrated QR code scanning for seamless event-entry tracking.

PROJECTS

Image Classification via CNNs | *PyTorch, Torchvision, NumPy, Matplotlib*

- Developed a CNN for multi-class classification on the CIFAR-10 dataset.
- Achieved 82% testing accuracy through architecture optimization and hyperparameter tuning.
- Applied advanced augmentation and normalization using torchvision to improve performance.

Custom Malware Detection using YARA | *YARA, Linux, Reverse Engineering Tools*

- Analyzed a Windows malware sample using static analysis to identify suspicious behaviors.
- Designed a custom YARA rule leveraging PE structure, anomalous timestamps, and string patterns.
- Engineered multi-condition detection logic and validated accuracy using binwalk and rabin2.

B-Minor Compiler | *C, Flex, Bison*

- Built a C-like language compiler using Flex and Bison for lexical and syntax analysis.
- Implemented name resolution and semantic type checking to ensure logic and type safety.
- Built an x86-64 code generator with stack management, control flow handling, and function calls.